

Finite-Memory Three-Mode Capacity

A Sharp Enclosure and Exterior-to-Fourth-Mode Recovery

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Abstract

RH-109 showed that the normalized spectral four-volume factors as $\nu_4 = \Lambda_{23}q_4$, where $q_4 = s_4/s_1$ and $\Lambda_{23} = (s_2/s_1)(s_3/s_1)$. This paper isolates and controls the capacity factor. If $\hat{s}_j$ are the singular values of a recent projected cross and the forgotten-memory operator radius is δ , we prove

$$\frac{(\hat{s}_2 - \delta)_+(\hat{s}_3 - \delta)_+}{(\hat{s}_1 + \delta)^2} \leq \Lambda_{23} \leq \frac{(\hat{s}_2 + \delta)(\hat{s}_3 + \delta)}{(\hat{s}_1 - \delta)_+^2}.$$

Dividing the RH-109 spectral-volume lower bound by the upper endpoint gives a rigorous capacity-aware fourth-mode certificate. On 360 archived threshold-update records its support count agrees exactly with the direct Weyl certificate: 113, 109, and 98 records at cutoffs $10^{-8}, 10^{-6}, 10^{-4}$. All 78 fine updates pass each cutoff, the minimum fine recovery efficiency is 0.9982343, and the widest fine capacity interval is 0.1761% of the true capacity. At fixed normalized volume ν , capacity obeys the sharp universal interval $\nu^{2/3} \leq \Lambda_{23} \leq 1$. Thus the factorization is now rigorous and numerically lossless on the archived chain, but an independent physical all-level capacity upper law remains open. No Stage A, Hilbert–Polya, zero-identification, or Riemann Hypothesis result is claimed.

1 Introduction

The source-seeded quotient route keeps four projected-cross directions when $q_4 = s_4/s_1$ exceeds an adaptive threshold [5, 6]. RH-108 gave a recent-memory Weyl certificate for this ratio, while RH-109 replaced the fourth mode by the coordinate-free spectral four-volume [3, 4]. The replacement loses the factor

$$\Lambda_{23}(K) = \frac{s_2(K)s_3(K)}{s_1(K)^2}, \quad \nu_4(K) = \Lambda_{23}(K)q_4(K). \quad (1)$$

The purpose of this paper is to determine whether that factor can be paid rigorously through the same finite-memory radius.

There are two questions. First, how stable is Λ_{23} under a small operator perturbation? Second, does dividing a volume lower bound by a capacity upper bound recover enough of the direct fourth-mode certificate to be useful? Both have exact finite-dimensional answers. The finite replay then shows that the composition loses less than two parts in a thousand on the fine scales.

2 Capacity and finite-memory perturbations

Let K and $\hat{K}$ map an r -dimensional packet into its complement, and assume

$$\left\| K - \hat{K} \right\|_2 \leq \delta. \quad (2)$$

Write $s_1 \geq \dots \geq s_r$ and $\widehat{s}_1 \geq \dots \geq \widehat{s}_r$ for their singular values. Set

$$\ell_j = (\widehat{s}_j - \delta)_+, \quad u_j = \widehat{s}_j + \delta. \quad (3)$$

Theorem 2.1 (Finite-memory three-mode capacity enclosure). *If $s_1 > 0$, then*

$$L_{23}^- := \frac{\ell_2 \ell_3}{u_1^2} \leq \Lambda_{23}(K) \leq L_{23}^+ := \frac{u_2 u_3}{\ell_1^2}, \quad (4)$$

where the upper endpoint is $+\infty$ when $\ell_1 = 0$.

Proof. Weyl's singular-value perturbation inequality gives $\ell_j \leq s_j \leq u_j$ for every j [1, 2]. Insert the lower numerator and upper denominator bounds for the left inequality, and the upper numerator and lower denominator bounds for the right inequality. $\square$

For normalized memory Gramians, RH-108 supplies

$$\delta_{t,m} = \eta^m \frac{1 - \eta^{t-m+1}}{1 - \eta}$$

for a depth- m recent cross. Hence Theorem 2.1 is a directly computable finite-memory enclosure.

3 Capacity-aware volume recovery

RH-109 proves the spectral-volume lower bound

$$B_4^{\text{vol}} = \frac{\ell_1 \ell_2 \ell_3 \ell_4}{u_1^4} \leq \nu_4(K). \quad (5)$$

Theorem 3.1 (Exterior-to-fourth-mode recovery). *Under the assumptions of Theorem 2.1,*

$$q_4(K) \geq B_4^{\text{cap}} := \begin{cases} B_4^{\text{vol}}/L_{23}^+, & 0 < L_{23}^+ < \infty, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

Moreover,

$$B_4^{\text{cap}} = \frac{\ell_4}{u_1} \left(\frac{\ell_1}{u_1} \right)^3 \frac{\ell_2}{u_2} \frac{\ell_3}{u_3} \leq \frac{\ell_4}{u_1}. \quad (7)$$

Proof. The exact identity (1) and $\Lambda_{23}(K) \leq L_{23}^+$ give $q_4 = \nu_4/\Lambda_{23} \geq B_4^{\text{vol}}/L_{23}^+$. Substitution of (5) and (4) gives (7). Every displayed efficiency factor is at most one. $\square$

The theorem explains both the strength and the limitation of the factorization. When $\delta/\widehat{s}_j$ is small, the extra factors are close to one and the direct Weyl ratio is recovered. If an intermediate singular value approaches the perturbation radius, the volume route loses that direction before the direct fourth-mode numerator necessarily does.

4 Sharp capacity range at fixed volume

Proposition 4.1 (Sharp fixed-volume capacity interval). *For $0 \leq \nu_4 \leq 1$,*

$$\nu_4^{2/3} \leq \Lambda_{23} \leq 1, \quad (8)$$

and both endpoints are attained.

Proof. RH-109 gives $\nu_4 \leq q_4 \leq \nu_4^{1/3}$. Since $\Lambda_{23} = \nu_4/q_4$, inversion yields (8). The spectrum $(1, 1, 1, \nu_4)$ gives $\Lambda_{23} = 1$, while $(1, \nu_4^{1/3}, \nu_4^{1/3}, \nu_4^{1/3})$ gives $\Lambda_{23} = \nu_4^{2/3}$. $\square$

The same endpoint spectra admit the trace-one source-seeded realization of RH-109 with fixed memory clock and diagonal blocks. Therefore no generic normalized-memory argument can replace the upper endpoint one by a smaller universal constant.

Table 1: Capacity-aware recovery counts versus direct Weyl counts.

cutoff	recovered	direct Weyl	fine recovered
10^{-8}	113/120	113/120	78/78
10^{-6}	109/120	109/120	78/78
10^{-4}	98/120	98/120	78/78

5 Sharpness and reduced realization

The capacity enclosure is sharp as a radius-only statement. Fix numbers $a > b > c > \delta$ and consider a recent diagonal cross whose first three singular values are a, b, c , together with an admissible fourth value. A perturbation acting negatively on the second and third right singular directions changes their values to $b - \delta$ and $c - \delta$, while a perturbation acting positively on the first direction changes the leading value to $a + \delta$. Thus the lower endpoint in (4) is attained in the diagonal model. Reversing the signs gives the upper endpoint. No improvement can follow from the single number $\|K - \hat{K}\|_2$ without additional alignment data.

The reduced-moment representation makes the same point intrinsically. If $C = K^*K$ has eigenvalues $\lambda_j = s_j^2$, then

$$\Lambda_{23}(K) = \frac{\sqrt{\lambda_2 \lambda_3}}{\lambda_1}, \quad q_4(K) = \frac{\sqrt{\lambda_4}}{\sqrt{\lambda_1}}. \quad (9)$$

For the recent memory, $C = M_2 - A^2$ is a packet-sized matrix. Hence the capacity interval can be evaluated without an ambient complement solve, but it still requires the second and third eigenvalues of that reduced matrix. The moment reduction changes the representation of the calculation; it does not create a lower bound on λ_2 or λ_3 .

Proposition 5.1 (Conditional capacity route). *Suppose a physical family admits, for all sufficiently fine levels, a volume lower bound $\nu_4 \geq v_t > 0$ and an independent capacity upper law $\Lambda_{23} \leq L_t$. Then*

$$q_4 \geq v_t / L_t. \quad (10)$$

In particular, $v_t \geq \tau L_t$ implies the no-quotient support gate $q_4 \geq \tau$.

Proof. Divide the identity $\nu_4 = \Lambda_{23} q_4$ by the positive upper bound L_t . $\square$

This proposition is deliberately conditional. It identifies the exact missing input for an all-level capacity route: one must obtain v_t and L_t from independent physical estimates, rather than read both from the same finite recent SVD.

6 Five-scale audit

We replay the two source-seeded channels at $\sigma \in \{0.16, 0.08, 0.04, 0.02, 0.01\}$, the three thresholds $10^{-8}, 10^{-6}, 10^{-4}$, memory ratio $\eta = 1/512$, and recent depth five. At each update the full cross is used only as a finite comparator. The certificate itself uses the recent spectrum and the guarded RH-108 tail.

All 360 capacity intervals contain the actual factor and every recovery implication is valid. On the fine scales

$$\begin{aligned} 5.23356 \times 10^{-5} &\leq \Lambda_{23} \leq 0.387636, \\ \max \frac{L_{23}^+ - L_{23}^-}{\Lambda_{23}} &= 1.76065 \times 10^{-3}, \\ \min \frac{B_4^{\text{cap}}}{\ell_4 / u_1} &= 0.9982343. \end{aligned}$$

Table 2: Observed capacity ranges for the primary threshold chain.

scale	minimum Λ_{23}	maximum Λ_{23}	updates
0.16	1.07×10^{-11}	1.85×10^{-3}	8
0.08	3.53×10^{-6}	1.87×10^{-2}	12
0.04	1.72×10^{-6}	2.90×10^{-1}	22
0.02	5.23×10^{-5}	3.44×10^{-1}	34
0.01	1.39×10^{-4}	3.88×10^{-1}	44

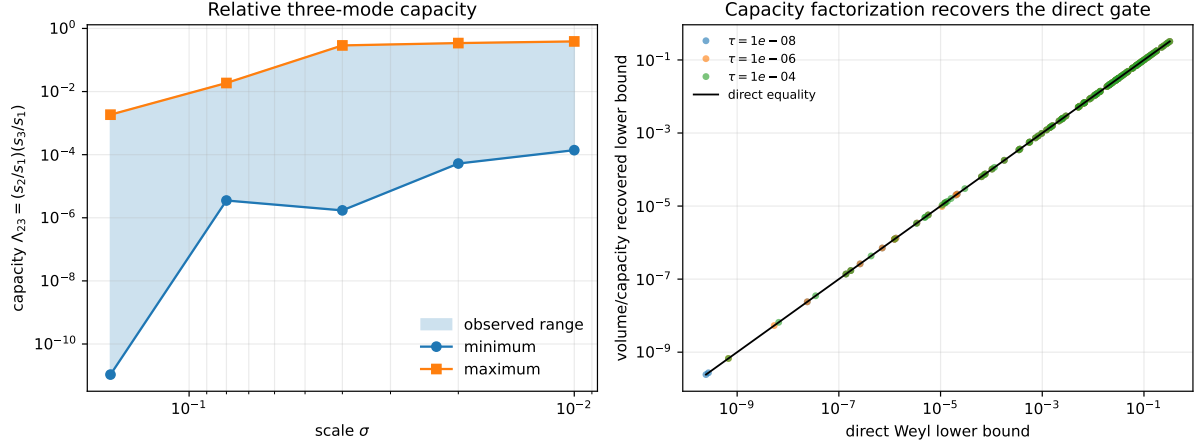


Figure 1: Left: the physical capacity range across scales. Right: the capacity-aware recovery lower bound lies almost on the direct Weyl lower bound; the remaining loss is exactly the product in (7).

Thus the volume/capacity decomposition reproduces the direct support gate to better than 0.18% on every fine record. The maximum relative enclosure width over all five scales is 0.0928 and occurs in a weak coarse branch.

7 Route consequence and claim boundary

RH-110 closes the algebraic composition

recent volume lower bound + recent capacity upper bound $\implies$ fourth-mode support.

It does not yet produce independent physical inputs: both factors are currently extracted from the same recent singular spectrum. An all-level argument must replace at least the capacity upper endpoint by a simpler source/observation, energy, or recurrence estimate, and must still prove a physical volume lower bound.

The exact claims are the capacity enclosure, recovery theorem, sharp fixed-volume interval, and finite five-scale validation. Open are an all-level capacity law, all-level exterior lower bound, unconditional fine-support separation, and Stage A. No Hilbert–Polya operator, zero identification, prime-power trace formula, or Riemann Hypothesis conclusion is asserted.

8 Reproducibility

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_capacity_audit.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_capacity_audit.py --smoke
```

```
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/make_figures.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/pytest -q -p no:cacheprovider
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
cp main.pdf finite-memory-three-mode-capacity.pdf
```

References

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